

CAMPUS CARBON

Universal Campus Carbon Footprint Assessment & Management Platform (v2.0)

Organization:	Christ University	Reporting Year:	2025
Campus Name:	Apex Institute of Science & Tech	Assessment Status:	Completed
Campus Type:	University / College	Total Population:	3,250 occupants
Location:	Bengaluru, Karnataka, None, India	Built-up Area:	80,937.0 m² (50.0 Acres)
Standard / Protocol:	GHG Protocol Corporate Standard / ISO 14064-1	Generated On:	20 September 2026, 17:04

1. GROSS / POSITIVE EMISSIONS	2. ELIGIBLE REDUCTIONS & REMOVALS	3. NET CARBON FOOTPRINT
886.99	-372.80	514.18
tCO2e	tCO2e	tCO2e

1. Executive Carbon Accounting Breakdown

Scope Classification	Activity Description	Emissions (tCO2e)	Share of Gross (%)
Scope 1 (Direct)	Stationary DG diesel, canteen LPG, campus fleet combustion, refrigerants	147.78	16.7%
Scope 2 (Indirect - Electricity)	Purchased grid electricity & EV charging consumption	716.00	80.7%
Scope 3 (Other Indirect)	Commuting, solid waste disposal, freshwater supply, wastewater STP	23.20	2.6%
Total Gross Emissions	All combined operational activity emissions	886.99	100.0%
Eligible Removals / Reductions	Tree carbon sequestration, verified EV savings, waste diversion	-372.80	-42.0%
Official Net Carbon Footprint	Gross Emissions – Eligible Reductions & Removals	514.18	58.0%

Gross Intensity / Person:	Net Intensity / Person:	Gross Intensity / Area:	Renewable Energy Share:	Waste Diversion Rate:
0.2729 tCO2e/person	0.1582 tCO2e/person	10.96 kgCO2e/m²	31.0%	100.0%

2. Building-wise Carbon Performance Leaderboard

Code	Building Name	Type	Area (m²)	Occ.	Gross (t)	Red. (t)	Net (t)	tCO2e/p	kg/m²
APX-01	Academic Wing 1	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-02	Academic Wing 2	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-03	Academic Wing 3	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-04	Academic Wing 4	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-05	Academic Wing 5	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-06	Academic Wing 6	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-07	Academic Wing 7	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-08	Academic Wing 8	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-09	Academic Wing 9	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0
APX-10	Academic Wing 10	Academic	8,094	325	0.00	0.00	0.00	0.00	0.0

3. Negative / Eligible Reduction Contributions Ledger

Category	Activity & Intervention	Methodology & Baseline	Status	Avoided / Sequestered
Solar Plant	Solar Clean Power Generation (Solar...	Grid Electricity Displacement (CEA India...	Eligible	-322.20 tCO2e
Gardening / Greenery	Biomass Sequestration by Campus Tre...	FSI Tree Biomass Growth Model & IPCC GPG...	Eligible	-33.00 tCO2e
Waste Diversion	Avoided Landfill Methane via Compos...	Avoided Landfill Methane Generation (IPC...	Eligible	-17.60 tCO2e

4. Prioritized Recommendations & Decarbonization Action Plan

Priority	Category	Intervention Action Title	Intervention Type	Est. Impact
High	Electricity	On-site Rooftop Solar PV Expansion & Sma...	Emission Reduction	-25%
High	Electricity	HVAC & Lighting Deep Energy Efficiency R...	Emission Reduction	-15%
Medium	Greenery	Afforestation & Miyawaki Native Urban Fo...	Carbon Sequestration / Removal	-8%
Low	Water	100% Treated Wastewater Recycling for La...	Emission Reduction	-5%

5. Accounting Boundaries & Anti-Double-Counting Disclosures

Methodological Boundary: This assessment is executed under the operational control consolidation approach in strict accordance with the GHG Protocol Corporate Accounting and Reporting Standard and ISO 14064-1:2018 specifications.

Anti-Double-Counting Safeguards:

1. On-site Solar Generation: Solar energy consumed directly on campus displaces grid power and is reflected exclusively in reduced Scope 2 emissions. It is NOT subtracted a second time as a separate negative carbon offset.
2. EV Fleet Accounting: Operational EV charging electricity is quantified under Scope 2 electricity. Avoided transport reductions are calculated separately against an explicit ICE baseline.
3. Carbon Sequestration: Tree and vegetation biomass sequestration is modeled based on Forest Survey of India & IPCC GPG factors (22.0 kg CO2/tree/year) and reported in the eligible removals ledger.